

Practical - 1

Practical Title: Project Initiation, Requirement Analysis, Sitemap, Wireframe, and GitHub Setup

Problem Definition: Initiate the semester-long StudentHub portal by identifying the problem scope, user roles, key modules, navigation flow, and minimum 10 pages. Create a sitemap, low-fidelity wireframe, project folder structure, README file, and GitHub repository.

A. Theory

1. What is URL and parts of URL?

Answer: A **URL** (Uniform Resource Locator) is the address of a unique resource on the internet. It is one of the key mechanisms used by browser to retrieve published resources, such as HTML pages, CSS documents, images, and so on. Scheme, Subdomain, Domain Name, Top-Level Domain, Port Number, Path, Query Parameters, Fragment .

(https://developer.mozilla.org/en-US/docs/Learn_web_development/Howto/Web_mechanics/What_is_a_URL)

2. How does HTML file process in web browser?

Answer: -You Write HTML Code

-You create an HTML file that contains the structure and content of your webpage — text, headings, images, links, forms, etc.

-You Open the File in a Browser

-When you double-click the file or open it through a web server, a browser like Chrome, Firefox, or Safari reads the code.

-The Browser Parses the HTML

-The browser scans your code from top to bottom. It starts with the `<!DOCTYPE html>` declaration, then reads sections.

-The Browser Builds the Page

-It displays content (like text, images, and videos) inside the

.

-It applies styles from CSS (if linked or embedded).

-It runs JavaScript for interactivity (if included).

Result: A Visual Web Page

What you see in the browser is a visual version of the HTML. The browser turns your code into a formatted page that looks clean and interactive — just like any website you visit.

<https://dev.to/ofodile/how-html-works-in-the-browser-25pa>

3. How will page navigation flow managed among all HTML pages?

Answer: - **Mapping your structure: page hierarchy, menus and links**

Before diving into specific components, it helps to state that **how users move through the information or navigate the pages of a website** depends on a coherent skeleton.

The following sections break down that skeleton into manageable parts.

- **Page hierarchy as the backbone**
- **Breadcrumb trails and path indicators**
- **Responsive behaviour across devices**
- **Internal linking for discoverability**
- **Measuring and mapping flows**
- **Designing for efficient movement**
- **Scroll behaviour and interactive elements**

<https://www.parallelhq.com/blog/how-users-move-through-information-or-navigate-pages-of-website>

4. How will GitHub commits be maintained after each practical?

Answer:

- git init
- git add .
- git commit -m "any commit message"
- git push origin main/master

<https://github.com/git-guides>

Implementation/ Output/Result

Project Requirements for StudentHub-Portal

Pages Included:

- index.html
- PAGES/home.html
- PAGES/about.html
- PAGES/admin.html
- PAGES/contact.html
- PAGES/dashboard.html
- PAGES/edit.html
- PAGES/events.html
- PAGES/faq.html
- PAGES/feedback.html
- PAGES/login.html
- PAGES/profile.html
- PAGES/register.html

Core Requirements:

- Static web application using HTML, CSS, and JavaScript
- Responsive layout for desktop and mobile screens
- Navigation across all pages via links or menu
- Login and registration forms on `PAGES/login.html` and `PAGES/register.html`
- User dashboard and profile pages for authenticated users

- Admin interface on `PAGES/admin.html`
- Contact page with message submission form
- Events listing page on `PAGES/events.html`
- FAQ, feedback, and about pages for user support and information
- Edit page to allow profile or content updates

B. Evidence of Work (Screenshots)

SiteMap:

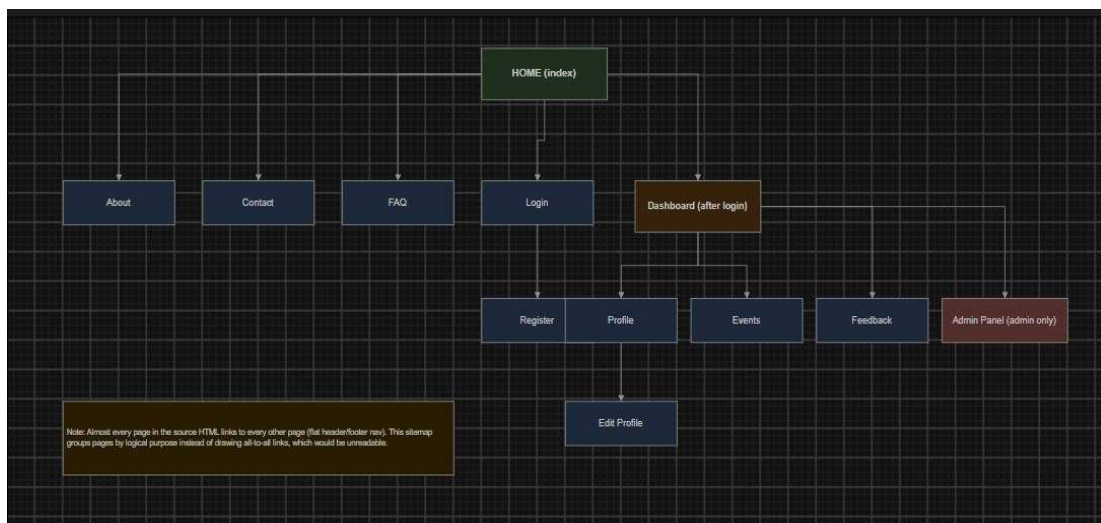


Figure 1.1

WireFrame:

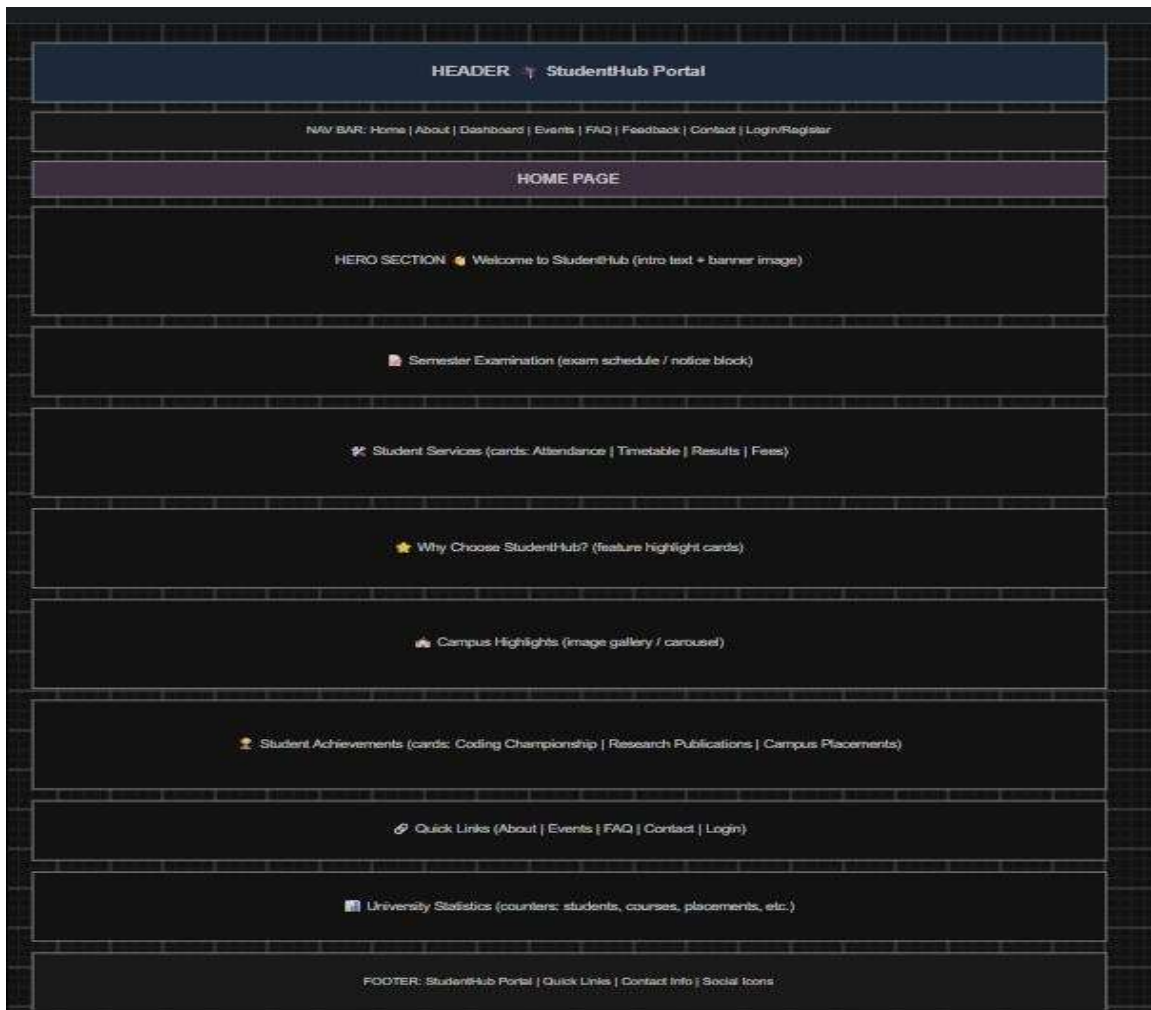


Figure 1.2(Home page)

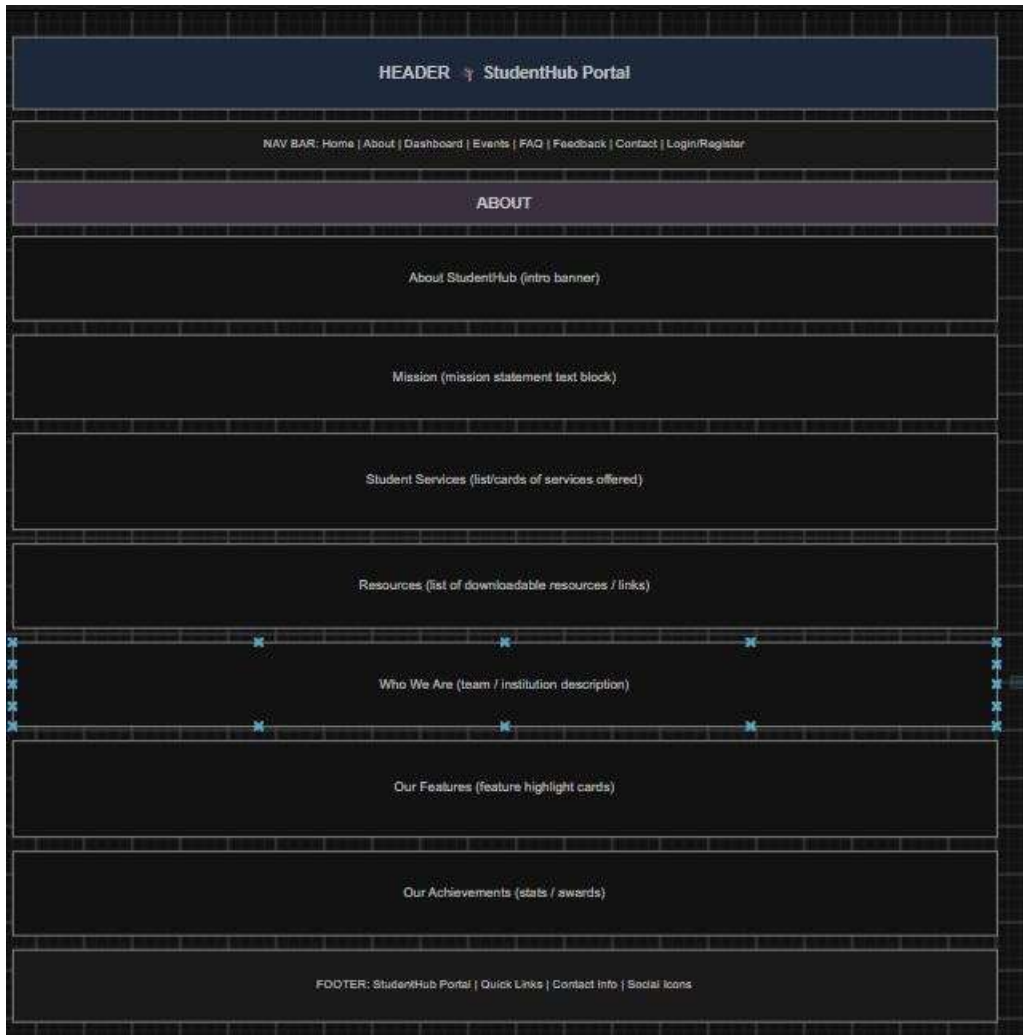
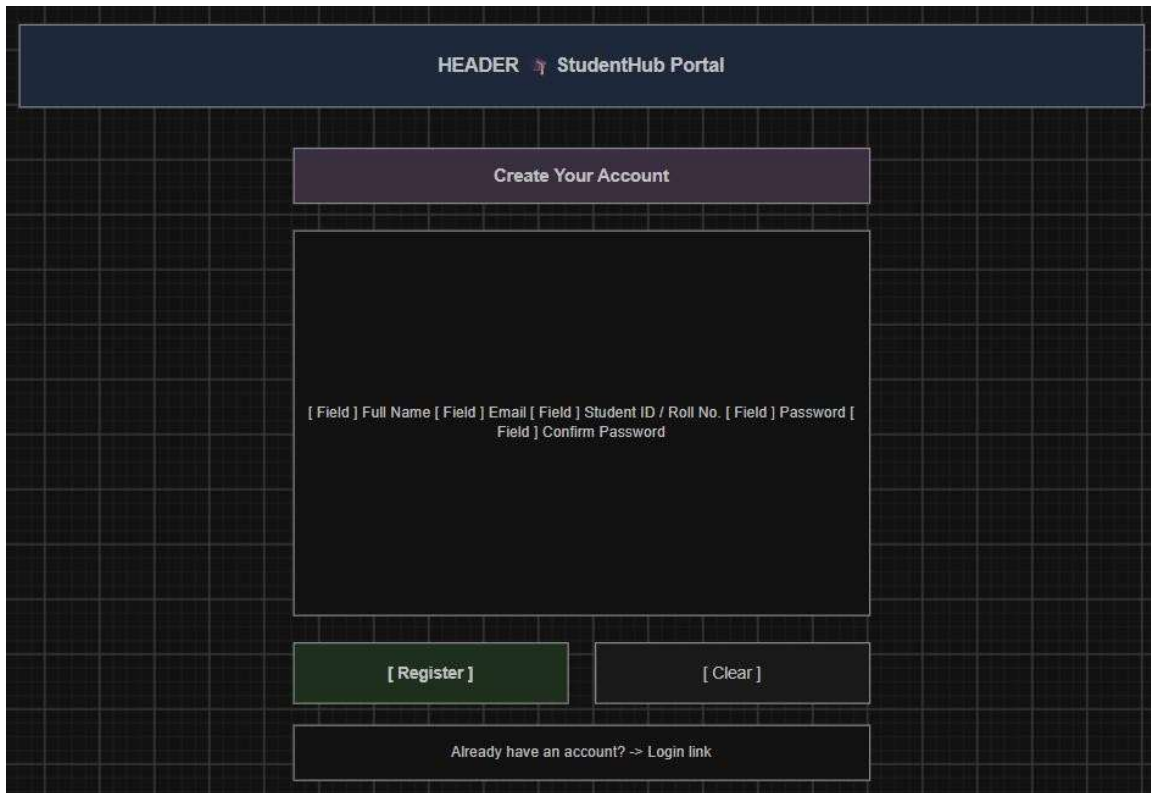


Figure 1.3(About page)



HEADER StudentHub Portal

Create Your Account

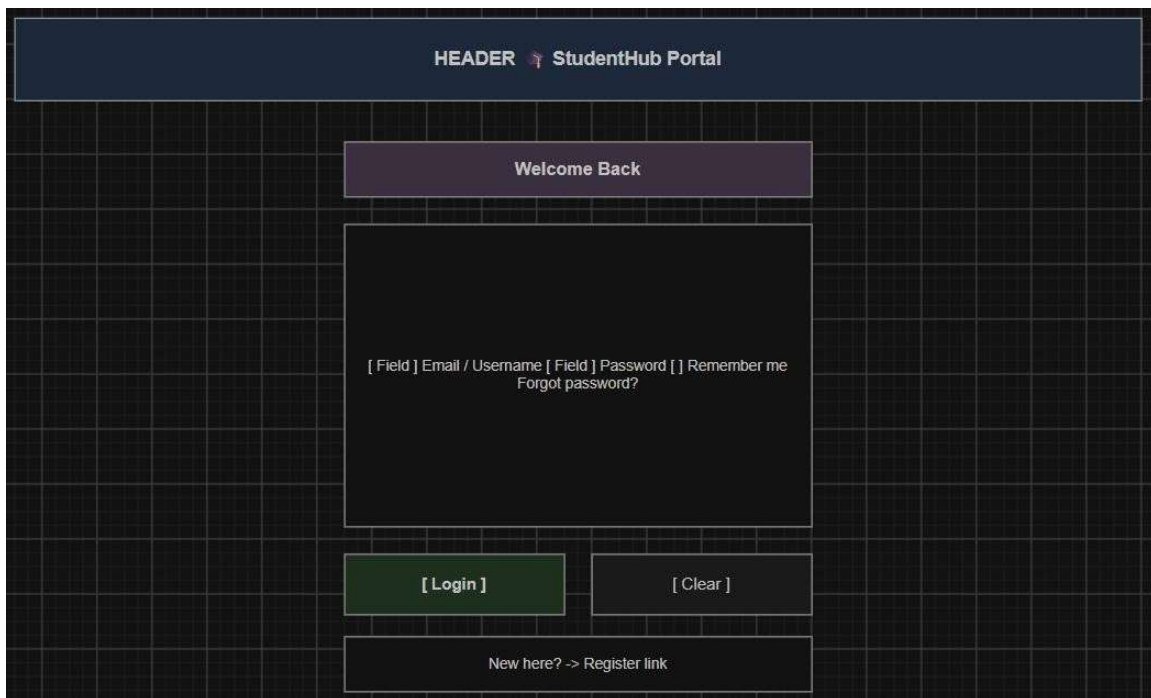
[Field] Full Name [Field] Email [Field] Student ID / Roll No. [Field] Password [Field] Confirm Password

[Register] [Clear]

Already have an account? -> Login link

The register page features a dark blue header with the text 'HEADER StudentHub Portal'. Below the header is a purple button labeled 'Create Your Account'. The main form area is a large white rectangle containing four input fields: 'Full Name', 'Email', 'Student ID / Roll No.', and 'Password', followed by a 'Confirm Password' field. Below the form are two buttons: a green 'Register' button and a grey 'Clear' button. At the bottom is a white box with the text 'Already have an account? -> Login link'.

Figure 1.4(Register page)



HEADER StudentHub Portal

Welcome Back

[Field] Email / Username [Field] Password [] Remember me
Forgot password?

[Login] [Clear]

New here? -> Register link

The login page features a dark blue header with the text 'HEADER StudentHub Portal'. Below the header is a purple button labeled 'Welcome Back'. The main form area is a large white rectangle containing two input fields: 'Email / Username' and 'Password', followed by a 'Remember me' checkbox and a 'Forgot password?' link. Below the form are two buttons: a green 'Login' button and a grey 'Clear' button. At the bottom is a white box with the text 'New here? -> Register link'.

Figure 1.5(Login page)

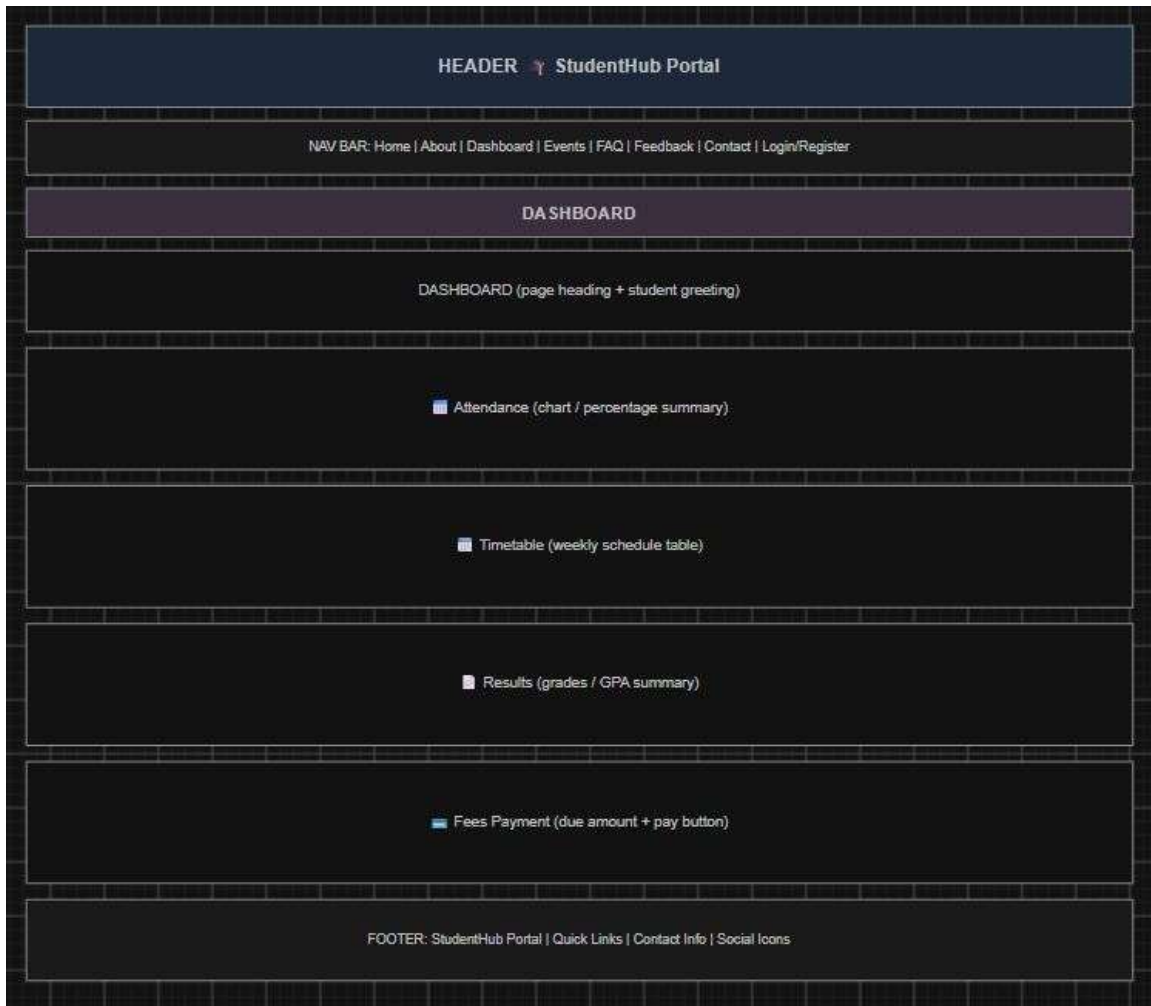


Figure 1.6(Dashboard page)



Figure 1.7(Event page)

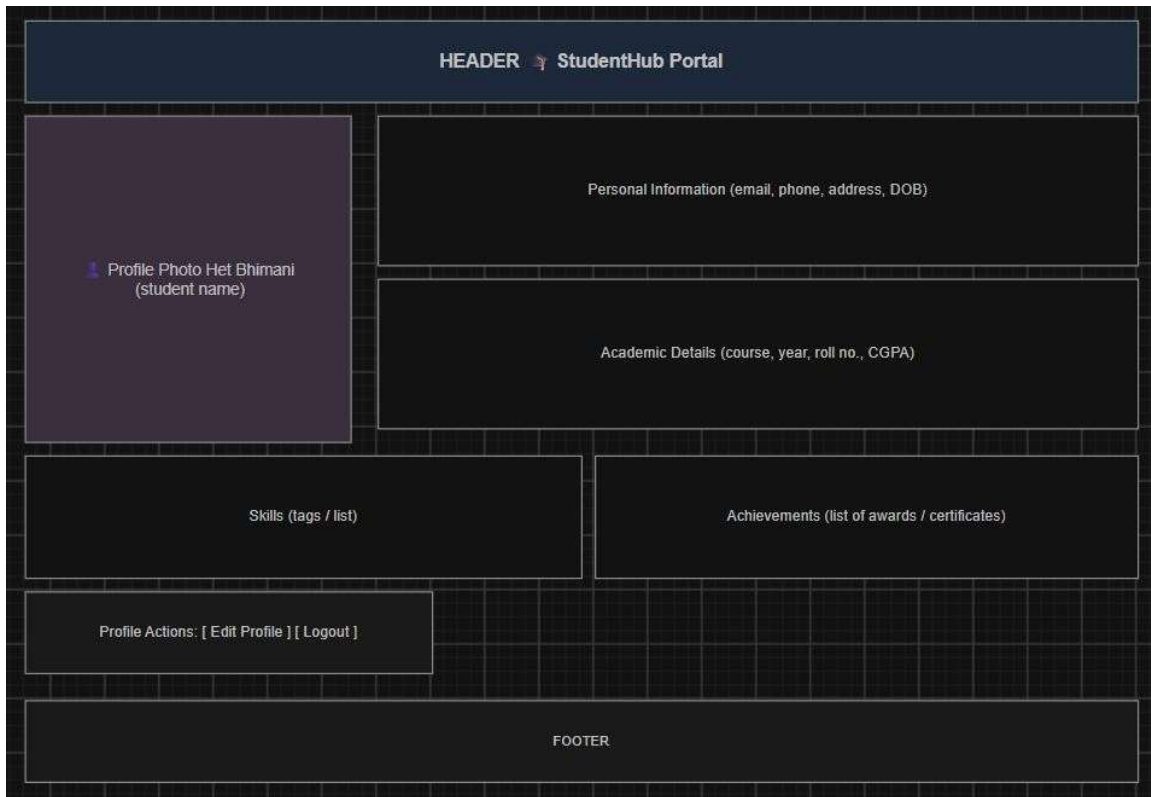


Figure 1.8(Profile page)

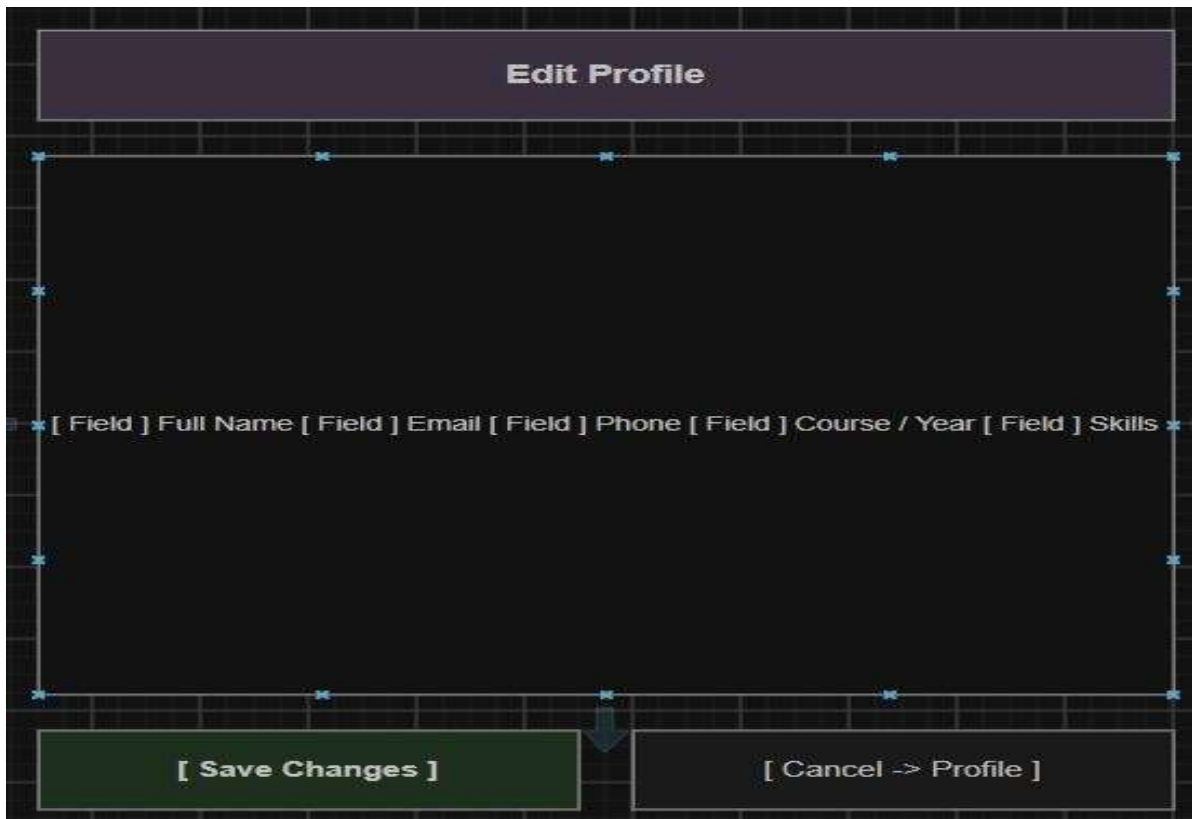


Figure 1.9(Edit page)

HEADER StudentHub Portal

NAV BAR: Home | About | Dashboard | Events | FAQ | Feedback | Contact | Login/Register

CONTACT

Contact Us (page heading)

Contact Information (address, phone, email, social icons)

Send Us a Message [Field] Name [Field] Email [Field] Message (textarea) [Send]

Office Hours (Mon-Fri timing table)

Email Support (support email + response time)

Find Us (embedded Google Map)

FOOTER: StudentHub Portal | Quick Links | Contact Info | Social Icons

Figure 1.10(Contact page)

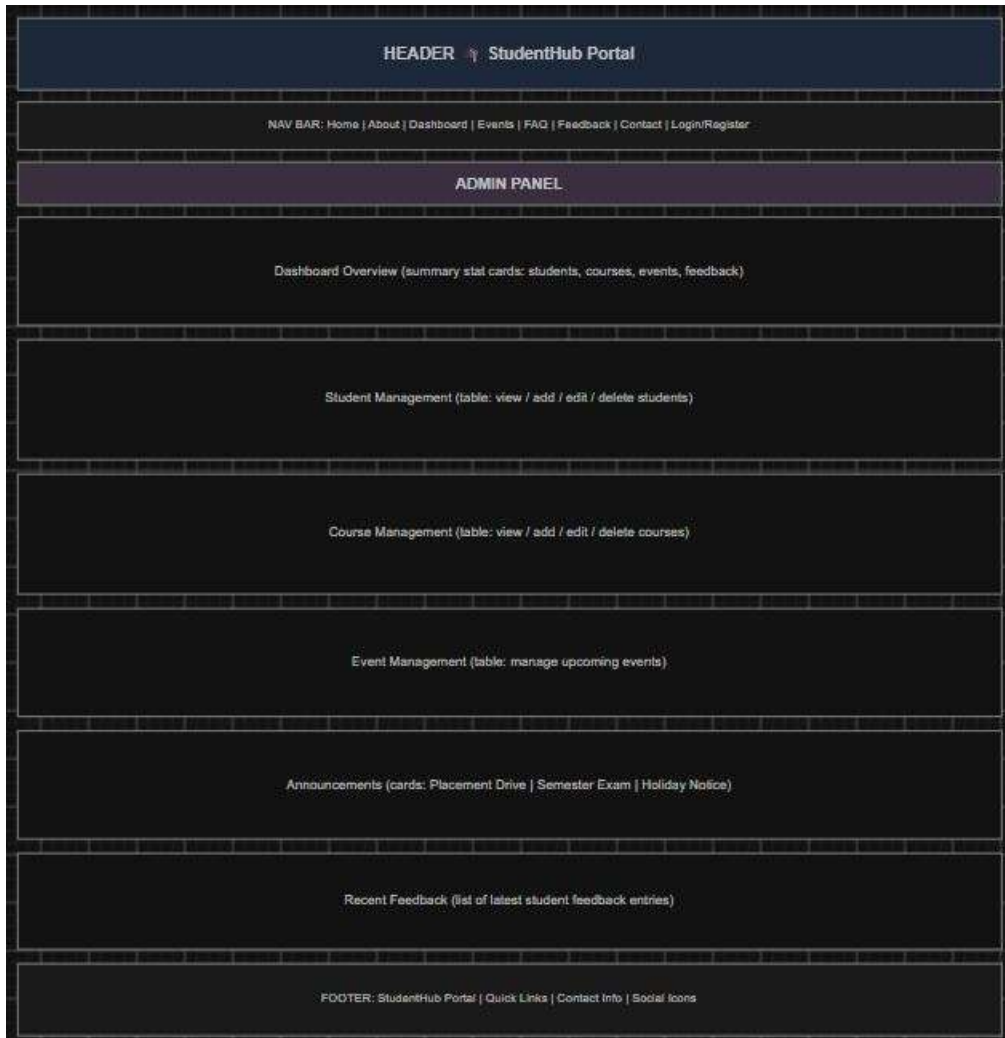


Figure 1.11(Admin page)

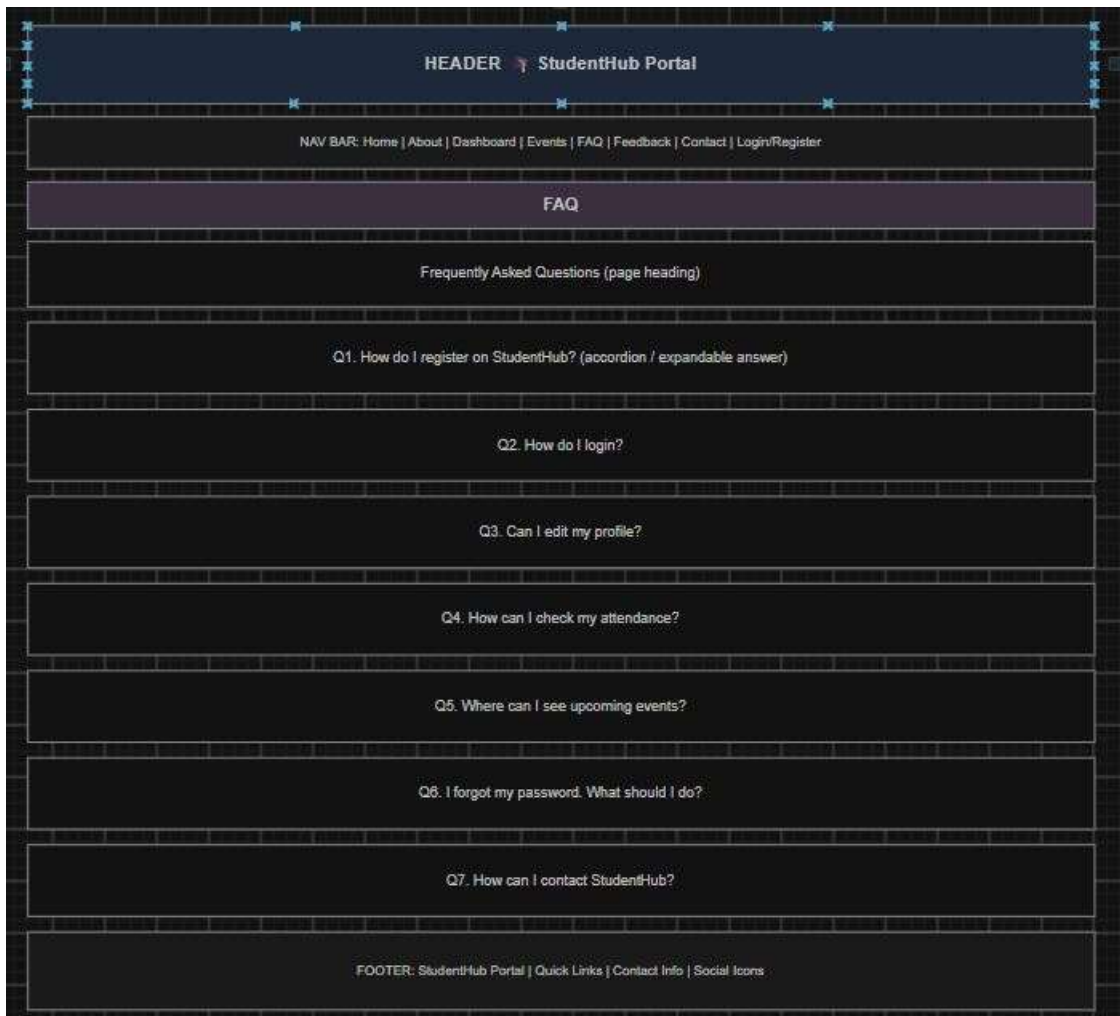


Figure 1.12(FAQ page)

The wireframe shows a dark-themed user interface. At the top is a dark blue header bar with the text 'HEADER' and a small icon followed by 'StudentHub Portal'. Below the header is a purple box labeled 'Student Hub Feedback Form'. Underneath this is a large dark box for 'Personal Information' containing labels for 'Name', 'Email', and 'Roll No.', each followed by a '[Field]' placeholder. Below that is another large dark box for 'Feedback Details' containing labels for 'Category (dropdown)', 'Rating (1-5)', and 'Comments (textarea)', each followed by a '[Field]' placeholder. At the bottom are two buttons: a green one labeled '[Submit]' and a dark one labeled '[Reset Form]'.

Figure 1.13(Feedback page)

C. GitHub Repository Link:

<https://github.com/yakshu205/StudentHub.git>

D. Challenges Faced:

First time when I start to think about requirement at that time I stuck . I thought that what type of pages we will make etc . But I start collecting info or requirement from my friend and then , I make sitemap and wireframe in tool like draw.io. I also faced difficulty when I make repos and push code and commit changes through command prompt , then I show a tutorial video of Git/GitHub version control , then I am good commit and etc.

E. Learning Outcomes:

I will be able to analyze requirements and prepare a structured project foundation before implementation and I am able to push , pull and commit my changes into my Github repo.

Faculty Remarks: (Keep this blank for faculty)